3)Arrays and Strings

1)

```
#include<stdio.h>
#include<string.h>

int main()
{
    char str1[20] = "Hello", str2[20] = " World";
    printf("%s\n", strcpy(str2, strcat(str1, str2)));
    return 0;
}
```

Output :Hello World.
Explanation: Step 1: char str1[20] = "Hello", str2[20] = " World"; The variable str1 and str2 is declared as an array of .characters and initialized with value "Hello" and " World" respectively
;(((Step 2: printf("%s\n", strcpy(str2, strcat(str1, str2
strcat(str1, str2)) it append the string str2 to str1. The result will be stored in str1. Therefore str1 contains "Hello <= ."World
.strcpy(str2, "Hello World") it copies the "Hello World" to the variable str2 <=
"Hence it prints "Hello World

.....

2)
which of the following function is used to find the first occurrence of a given string in another string?

Output: strstr().
Explanation: Declaration: char *strstr(const char *s1, const char *s2);

```
int main(void)
{
  char *str1 = "Mohamed", *str2 = "am", *ptr;
  ptr = strstr(str1, str2);
  printf("The substring is: %s\n", ptr);
  return 0;
}
```

.....

3) What will be the output of the program if the array begins at 65472 and each integer occupies 2 bytes?

```
#include<stdio.h>

int main()
{
    int a[3][4] = {1, 2, 3, 4, 4, 3, 2, 1, 7, 8, 9, 0};
    printf("%u, %u\n", a+1, &a+1);
    return 0;
}
```

Output:65480 65496.
Explanation: For a two-dimensional array like a reference to array has type "pointer to array of 4 ints". Therefore, a+1 is pointing to the memory location of first element of the second row in array a. Hence 65472 + (4 ints * 2 bytes) = 65480
Then, &a has type "pointer to array of 3 arrays of 4 ints", totally 12 ints. Therefore, &a+1 denotes "12 ints * 2 bytes * 1 = 24 bytes".

.....

4)

```
#include<stdio.h>
void main()
{
   char arr[20]="MysticRiver";
   printf("%d",sizeof(arr));
}
```

A- 20
B- 11
C- 12
D- 22
E- 24

Output:20.
Explanation:Notice size between brackets of array.

.....

5) assume double=8 byte ,int=2 byte,char=1byte.

```
#include<stdio.h>
void main(){
    long double a;
    signed char b;
    int arr[sizeof(!a+b)];
    printf("%d",sizeof(arr));
}
```

Output:4.
Explanation:!a will convert to int so int+char=int(2 Bytes), so size of array is 2*2=4Bytes.

.....

6)

```
#include<stdio.h>

int main()
{
    printf(5+"Good Morning\n");
    return 0;
}
```

Output:Morning.
Explanation:it skips the first 5 characters and print the given string.

.....

7)

```c
#include<stdio.h>

int main()
{
    char *names[] = { "Suresh", "Siva", "Sona", "Baiju", "Ritu"};
    int i;
    char *t;
    t = names[3];
    names[3] = names[4];
    names[4] = t;
    for(i=0; i<=4; i++)
        printf("%s,", names[i]);
    return 0;
}
```

```
Output: Suresh, Siva, Sona, Ritu, Baiju.
Explanation:
Step 1: char *names[] = { "Suresh", "Siva", "Sona", "Baiju", "Ritu"}; The
.variable names is declared as an pointer to a array of strings
.Step 2: int i; The variable i is declared as an integer type
.Step 3: char *t; The variable t is declared as pointer to a string
Step 4: t = names[3]; names[3] = names[4]; names[4] = t; These statements the
.swaps the 4 and 5 element of the array names
Step 5: for(i=0; i<=4; i++) printf("%s,", names[i]); These statement prints
.the all the value of the array names
"Hence the output of the program is "Suresh, Siva, Sona, Ritu, Baiju
```

……

8) What will be the output of the program if the array begins 1200 in memory? *

```c
#include<stdio.h>

int main()
{
    int arr[]={2, 3, 4, 1, 6};
    printf("%u, %u, %u\n", arr, &arr[0], &arr);
    return 0;
}
```

```
Output:1200 1200 1200.
Explanation:
Step 1: int arr[]={2, 3, 4, 1, 6}; The variable arr is declared as an integer
.array and initialized
,Step 2: printf("%u, %u, %u\n", arr, &arr[0], &arr); Here
.The base address of the array is 1200
.arr, &arr is pointing to the base address of the array arr <=
arr[0] is pointing to the address of the first element array arr. (ie. base& <=
(address
Hence the output of the program is 1200, 1200, 1200
```

……

9)

```
#include<stdio.h>
#include<string.h>

int main()
{
    printf("%d\n", strlen("123456"));
    return 0;
}
```

Output:6.
Explanation: count number of characters until null terminator.

.....

10)

```
#include<stdio.h>
#include<string.h>

int main()
{
    static char str1[] = "dills";
    static char str2[20];
    static char str3[] = "Daffo";
    int i;
    i = strcmp(strcat(str3, strcpy(str2, str1)), "Daffodills");
    printf("%d\n", i);
    return 0;
}
```

Output: 0
Explanation: if two strings is equal so strcmp return zero.

.....

11)

```
#include<stdio.h>
#include<string.h>

int main()
{
    char sentence[80];
    int i;
    printf("Enter a line of text\n");
    gets(sentence);
    for(i=strlen(sentence)-1; i >=0; i--)
        putchar(sentence[i]);
    return 0;
}
```

Output:The sentence will get printed in reverse order
Explanation: 'i'is initialized with the last element in the string ad decrement till reach the first character in string.

.....

12)

```
#include<stdio.h>

int main()
{
    char str[25] = "IndiaBIX";
    printf("%s\n", &str+2);
    return 0;
}
```

```
Output: Garbage value.
Explanation: Step 1: char str[25] = "IndiaBIX"; The variable str is declared as
."an array of characteres and initialized with a string "IndiaBIX
;(Step 2: printf("%s\n", &str+2
In the printf statement %s is string format specifier tells the compiler to  <=
print the string in the memory of &str+2
str is a location of string "IndiaBIX". Therefore &str+2 is another memory & <=
.location
.Hence it prints the Garbage value
```

……

13)

```
#include<stdio.h>
#include<string.h>

int main()
{
    char str[] = "India\0\BIX\0";
    printf("%d\n", strlen(str));
    return 0;
}
```

```
Output:5
Explanation:
The function strlen returns the number of characters int the given string
Therefore, strlen(str) becomes strlen("India") contains 5 characters. A string
'is a collection of characters terminated by '\0
"The output of the program is "5
```

……

14)

```
#include<stdio.h>
void main()
{
   int const SIZE=5;
   int expr;
   double value[SIZE]={2.0,4.0,6.0,8.0,10.0};
   expr=1|2|3|4;
   printf("%f",value[expr]);
}
A-2.000000
B-4.000000
C-6.000000
D-8.000000
E- Compilation error
```

```
Output: Compilation error.
Explanation: we must use constant size for array.
```

15)

Which of the following function is more appropriate for reading in a multi-word string?

Output: gets().

Explanation: to read multi-word we use gets() or fgets(), fgets is safer than gets.

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16)

What will be the output of the program?

```
#include<stdio.h>
void main()
{
    int a[5] = {5, 1, 15, 20, 25};
    int i, j, m;
    i = ++a[1];
    j = a[1]++;
    m = a[i++];
    printf("%d, %d, %d", i, j, m);
}
```

Output:3,2,15.

Explanation: Step 1: int a[5] = {5, 1, 15, 20, 25}; The variable arr is declared as an integer array with a size of 5 and it is initialized to

. a[0] = 5, a[1] = 1, a[2] = 15, a[3] = 20, a[4] = 25

.Step 2: int i, j, m; The variable i,j,m are declared as an integer type

Step 3: i = ++a[1]; becomes i = ++1; Hence i = 2 and a[1] = 2

.Step 4: j = a[1]++; becomes j = 2++; Hence j = 2 and a[1] = 3

Step 5: m = a[i++]; becomes m = a[2]; Hence m = 15 and i is incremented by (1(i++ means 2++ so i=3

Step 6: printf("%d, %d, %d", i, j, m); It prints the value of the variables i, j, m

Hence the output of the program is 3, 2, 15

.....

17)

```
#include<stdio.h>

int main()
{
    int arr[1]={10};
    printf("%d\n", 0[arr]);
    return 0;
}
```

Output: 10.

Explanation:0[arr]=arr[0].(later in pointer).

18)

What will be the output of the following program in 16 bit platform assuming that 1022 is memory address of the string "Hello1" ?

```
#include<stdio.h>

int main()
{
    printf("%u %s\n", &"Hello1", &"Hello2");
    return 0;
}
```

Output: 1022 Hello2.
Explanation: In printf("%u %s\n", &"Hello", &"Hello");.
The %u format specifier tells the compiler to print the memory address of the "Hello1".
The %s format specifier tells the compiler to print the string "Hello2".
Hence the output of the program is "1022 Hello2".

……

19)

```
#include<stdio.h>

int main()
{
    char p[] = "%d\n";
    p[1] = 'c';
    printf(p, 65);
    return 0;
}
```

Output: A.
Explanation: Step 1: char p[] = "%d\n"; The variable p is declared as an array of characters and initialized with string "%d"
Step 2: p[1] = 'c'; Here, we overwrite the second element of array p by 'c'. So array p becomes "%c"
Step 3: printf(p, 65); becomes printf("%c", 65),Therefore it prints the ASCII 'value of 65. The output is 'A

……

20)

```
#include<stdio.h>
#include<string.h>

int main()
{
    char str[] = "India\0\BIX\0";
    printf("%s\n", str);
    return 0;
}
```

Output :India.
Explanation:str will print till found null terminated('\0').

21)

```
#include<stdio.h>
void main()
{
   char arr[11]="The African Queen";
   printf("%s",arr)
}
```

A- The African Queen
B- The
C- Queen
D- null
E- Compilation error

Output: Warning and print Garbage value.
Explanation: Initializer string for array is so long.

.....

22)

How will you print \n on the screen? *

```
Printf("\\n");
```

.....

23)

```
#include<stdio.h>

int main()
{
    char str1[] = "Hello";
    char str2[] = "Hello";
    if(str1 == str2)
        printf("Equal\n");
    else
        printf("Unequal\n");
    return 0;
}
```

Output: Unequal.
Explanation: Step 1: char str1[] = "Hello"; The variable str1 is declared as an array of characters .and initialized with a string "Hello"
Step 2: char str2[] = "Hello"; The variable str2 is declared as an array of characters and initialized with a string "Hello"
.We have use strcmp(s1,s2) function to compare strings
Step 3: if(str1 == str2) here the address of str1 and str2 are compared. The address of both .variable is not same. Hence the if condition is failed
Step 4: At the else part it prints "Unequal"

.....

24)

What will happen if in a C program you assign a value to an array element whose subscript exceeds the size of array?
A- The element will be set to 0.
B- The compiler would report an error.
C- The program may crash if some important data gets overwritten.
D- The array size would appropriately grow.

Output: The program may crash if some important data gets overwritten.
Explanation: If the index of the array size is exceeded, the program will crash. Hence "option c" is the correct answer. But the modern compilers will take care of this kind of errors.
Since C is a compiler dependent language, it may give different outputs at .different platforms. We have given the Turbo-C Compiler (Windows) output Please try the above programs in Windows (Turbo-C Compiler) and Linux (GCC .Compiler), you will understand the difference better

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25)Assume that int is 2 Bytes.

```
#include<stdio.h>
void main(){
    int arr[][3]={{1,2},{3,4,5},{5}};
    printf("%d %d %d",sizeof(arr),arr[0][2],arr[1][2]);
}
```

Output: 18 0 5.
Explanation: we have 3 rows and 3 cols so array size is 3*3*sizeof(int)=18, arr[0][2]=0(the compiler put it) and arr[1][2]=5.

……

26)

```
#include<stdio.h>
void main()
{
   char arr[7]="Network";
   printf("%s",arr);
  }
```

A- Network
B- N
C- network
D- Garbage value
E- Compiler error

Output: Garbage Value.
Explanation:As size of array is 7 and number of elements is 8.

……

27)

```
void main()
{
    char a1[3] = "abc";
    char a2[8] = "abcdefgh";
    char b1[]  = "abc";
    char b2[]  = "abcdefgh";
    printf("a1:%s is size: %d\n",a1,sizeof(a1));
    printf("a2:%s is size: %d\n",a2,sizeof(a2));
    printf("b1:%s is size: %d\n",b1,sizeof(b1));
    printf("b2:%s is size: %d\n",b2,sizeof(b2));
}
```

Output:3 8 4 9.

.....

28)

Assume we defined an array as int arr[20], what will happen in if I write arr[20]?

PICK ONE OF THE CHOICES

- Warning
- Compilation error
- Linking error

Output: Compilation error.
Explanation: to define array we must put data type of array, so we can't write arr[20].

.....

29)

If the two strings are identical, then strcmp() function returns ?

Output:0.

Explanation: strcmp returns zero if two strings are identical, returns '1' if str1>str2. And returns '-1' if str1<str2.

.....

30)

What will be the output of following program code?

```
#include <stdio.h>
int main(void)
{
    char p;
    char buf[10] = {1, 2, 3, 4, 5, 6,
 9, 8};
    p = (buf + 1)[5];
    printf("%d", p);
    return 0;
}
```

Output:9.
Explanation:(buff+1)[5]=buff[6](later in pointers).

……

31)

An array elements are always stored in _____ memory locations.

A. Sequential

B. Random

C. Sequential and Random

D. None of the above

Output:Sequential.

……

32)

```
#include<stdio.h>

int main()
{
    printf("India", "BIX\n");
    return 0;
}
```

Output:India.
Explanation: `printf("India", "BIX\n");` It prints "India". Because , (comma) operator has Left to Right associativity. After printing "India", the statement got terminated.

33)

The library function used to find the last occurrence of a character in a string is ?

Output: strrchr().

Explanation: **Declaration**: `char *strrchr(const char *s, int c);`

```
int main(void)

{

  char text[] = "I learn through IndiaBIX.com";

  char *ptr, c = 'i';

  ptr = strrchr(text, c);

  if (ptr)

    printf("The position of '%c' is: %d\n", c, ptr-text);

  else

    printf("The character was not found\n");

  return 0;

}
```

.....

34)

n C, if you pass an array as an argument to a function, what actually gets passed?

Output:Base address of the array.
Explanation: Later in pointers(:.D)

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35)

```
#include<stdio.h>
void main(){
   char data[2][3][2]={0,1,2,3,4,5,6,7,8,9,10,11};
   printf("%d",data[0][2][1]);
}
```

A- 5
B- 6
C- 7
D- 8
E- Compiler error

Output:5.

36)

```c
#include<stdio.h>
#include<string.h>

int main()
{
    printf("%c\n", "abcdefgh"[4]);
    return 0;
}
```

Output:e.
Explanation: consider str="abcdefgh", and we want to print str[4], so will print e.

……

37)

```c
#include<stdio.h>

int main()
{
    int i;
    char a[] = "\0";
    if(printf("%s", a))
        printf("The string is not empty\n");
    else
        printf("The string is empty\n");
    return 0;
}
```

Output:The string is empty.
Explanation:printf will return zero so else will execute.

……

38)

```c
void main()
{
    int a;
    int b;
    for (a = 0,b = 0;a < 10,b < 5;a++,b++)
    {
        printf("%d %d\n",a,b);
    }
}
```

Output: 0 0 \n 1 1 \n 2 2 \n 3 3\n 4 4.
Explanation: comma here is separator so condition will be b<5.

39)

What is right way to Initialize array?

A. int num[6] = { 2, 4, 12, 5, 45, 5 };
B. int n{} = { 2, 4, 12, 5, 45, 5 };
C. int n{6} = { 2, 4, 12 };
D. int n(6) = { 2, 4, 12, 5, 45, 5 };

```
Output: Answer A.
```

……

40)

```c
#include<stdio.h>
void main(){
    int xxx[10]={5};
    printf("%d %d",xxx[1],xxx[9]);
}
```

```
Output: 0 0.
Explanation:the compiler will initialize the rest of elements
by zeros.
```